Dirichlet: infinitely many rational approximations with

$$\left|x - \frac{p}{q}\right| < \frac{1}{q^2}.$$

Khintchine: let $\psi : \mathbb{N} \to \mathbb{R}_+$ be non-increasing. Then for almost all $x \in \mathbb{R}$, the inequality

$$\left|x - \frac{p}{q}\right| < \psi(q)$$

has infinitely (resp. finitely) many solutions according as the sum $\sum_{q \geq 1} \psi(q)q$ is infinite (resp. finite).

Schmidt, 1960: Set

$$N_\psi(x, T) := \# \left\{(p, q) \in \mathbb{Z} \times \mathbb{N} : (*) \text{ holds, } q < T\right\}.$$

Then for almost all $x \in \mathbb{R}$, we have

$$N_\psi(x, T) \sim 2 \sum_{q=1}^{T} \psi(q)q \text{ as } T \to \infty.$$

Now let's say a bit about *flag varieties*. More generally, let G be a semisimple $\mathbb{Q}$-group, and P a parabolic $\mathbb{Q}$-subgroup. We set $X := G/P$. We choose a maximal compact subgroup K. We also need a height function $H_\chi : X(\mathbb{Q}) \to \mathbb{R}_+$. We also need a K-invariant Riemannian distance function $d(\cdot, \cdot)$. We denote by σ the corresponding K-invariance Riemannian probability measure, given by the normalized volume form.

For example, we can take

$$G = \mathrm{SL}_2(\mathbb{R}) > P = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}, \quad K = \mathrm{SO}(2),$$

so that $X = \mathbb{P}^1$, and

$$H : \mathbb{P}^1(\mathbb{Q}) \to \mathbb{R}_+$$

given as follows: given $v \in \mathbb{P}^1(\mathbb{Q})$, we can represent it by a primitive lattice point $\vec{v} \in \mathbb{Z}^2_{\text{prim}}$, and we then define the height to be the norm of that vector:

$$H(v) := \|\vec{v}\|.$$

We can picture this in terms of shortest integer vectors along rational lines.

Set

$$N_\tau(x, T) = \# \left\{v \in X(\mathbb{Q}) : d(x, v < H_\chi(v)^{-\tau}),\, H_\chi(v) < T\right\},$$

$$\mathcal{E}(x, T) = \left\{\vec{v} \in \mathbb{R}^2 - \{0\} : \|\vec{v}\| < T,\, d(x, [\vec{v}]) < \|\vec{v}\|^{-\tau}\right\}.$$

In the example of interest,

$$N_\tau(x, T) = \#(\mathbb{Z}^2_{\text{prim}} \cap \mathcal{E}(x, T)) = \sum_{v \in \mathbb{Z}^2_{\text{prim}}} 1_{\mathcal{E}(x,T)}(v).$$

We form the dyadic region

$$\mathcal{F}(x, T) := \mathcal{E}(x, T) - \mathcal{E}(x, T/2).$$

We can apply a flow, such as $a_y := \operatorname{diag}(y^{-1}, y)$, to this region. We choose y_T so that the flowed region, $\mathcal{B}(T)$, is nice and round. Then

$$\begin{aligned}\#(\mathbb{Z}^2_{\text{prim}} \cap \mathcal{F}(x,T)) &= \#\left(k_X \mathbb{Z}^2_{\text{prim}} \cap \mathcal{F}(T)\right)\\ &= \#\left(a_{y_T} k_X \mathbb{Z}^2_{\text{prim}} \cap \mathcal{B}(T)\right)\\ &\approx \operatorname{vol}(\mathcal{B}(T)).\end{aligned}$$

Here $\mathcal{B}(T)$ "looks like a square", while k_X is typically not "too distorted".

Lemma 1. There exists $\varepsilon > 0$ such that for all large T, and all $g \in G$ such that $\operatorname{height}(g) \ll T^{\varepsilon}$, *where*

$$\operatorname{height}(g) := \left(\min_{v \in \mathbb{Z}^2 - 0} \|gv\|\right)^{-1},$$

we have

$$\#\left(g\mathbb{Z}^2_{\text{prim}} \cap \mathcal{B}(T)\right) = \operatorname{vol}(\mathcal{B}(T))\left(1 + \mathrm{O}(T^{-\varepsilon})\right).$$

Proof sketch. This is a counting argument due to (or at least inspired by) Eskin–McMullen.

Let's first notice that what we want to estimate is, as we have said before, the primitive Siegel transform

$$F_T(g) := \#(g\mathbb{Z}^2_{\text{prim}} \cap \mathcal{B}(T)) = \sum_{v \in \mathbb{Z}^2_{\text{prim}}} 1_{\mathcal{B}(T)}(gv) = \sum_{\gamma \in \Gamma/\Gamma_\infty} 1_{\mathcal{B}(T)}(g\gamma e_1).$$

Here $\Gamma_\infty = \Gamma \cap U$, where $U := \left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right)$. Now we unfold:

$$\begin{aligned}\langle F_T, \phi\rangle &= \int_{G/\Gamma} F_T(g)\phi(g)\, d\mu(g) = \int_{G/\Gamma} \sum_{\gamma \in \Gamma/\Gamma_\infty} 1_{\mathcal{B}(T)}(g\gamma e_1)\phi(g)\, d\mu\\ &= \int_{G/U} \int_{U/\Gamma_\infty} 1_{\mathcal{B}(T)}(gue_1)\phi(gu)\, du\, dm\\ &= \int_{\mathbb{R}^2} 1_{\mathcal{B}(T)}(ka_y e_1) \int_{U/\Gamma_\infty} \phi(ka_y u)\, du\, dm.\end{aligned}$$

Now we use the equidistribution result of Sarnak (1981) to see that there exists $\alpha > 0$ so that for all $f \in C_c^\infty(G/\Gamma)$, we have

$$\left| \int_{U/\Gamma_\infty} \phi(ka_y u)\, du - \int_{G/\Gamma} \phi\, d\mu \right| \ll y^{-\alpha} S(\phi).$$

This yields the required asymptotic. The well-roundedness allows us to go from an average counting result to an individual counting result. Main term: $f = 1_{\mathcal{B}(T)}$,

$$\operatorname{vol}(\mathcal{B}(T)) = \int_{\mathbb{R}^2} f\, dm = \int_{G/\Gamma} \hat{f}\, d\mu.$$

Variance:

$$\int_{G/\Gamma} \left|\hat{f} - \operatorname{vol}(\mathcal{B}(T))\right|^2 d\mu = \int_{G/\Gamma} \left|\hat{f}\right|^2 d\mu - \operatorname{vol}(\mathcal{B}(T))^2.$$

The integral on the right hand side may be written

$$\langle \hat{f}, \hat{f}\rangle = \int_{\mathbb{R}^2} 1_{\mathcal{B}(T)}(ka_y e_1) \int_{U/\Gamma_\infty} E(ka_y u, f)\, du\, dm.$$

This last integral is a constant term of an Eisenstein series. It describes second moments of Siegel transforms. One obtains in this way a bound on the variance. One can also argue and get an asymptotic estimate. $\square$

Okay, here's our result:

Theorem 2 (Pfitscher 2024)**.** *Suppose that P is a* maximal *parabolic subgroup of any semisimple algebraic $\mathbb{Q}$-group G. Let $\tau \geq 0$ be arbitrary. Write $d := \dim X$, let β_χ denote the Diophantine exponent, and define*

$$\Psi(T) := \begin{cases} T^{(\beta_\chi - \tau)d} & \text{if } \tau < \beta_\chi, \\ \log T & \text{if } \tau = \beta_\chi. \end{cases}$$

Then there exists $c > 0$ and $\varepsilon > 0$ such that for almost every $x \in X$, the counting function is given by

$$\mathcal{N}_\tau(x, T) = c\Psi(T)\left(1 + \mathrm{O}_X(\Psi(T)^{-\varepsilon})\right).$$

Remark 3. For $\mathbb{R}^n$, you can prove that the second moment of the Siegel transform is given as follows:

$$\int_{G/\Gamma} |\hat{f}|^2 \, d\mu - \left(\int_{\mathbb{R}^n} f \, dm\right)^2 = \int_{\mathbb{R}} (|f|^2 + f \cdot f^-) \, dm \ll \mathrm{vol}(\mathcal{B}(T)),$$

where

$$f^-(v) = f(-v).$$

This estimate suffices to get the conclusion of the theorem (for an arbitrary decreasing Ψ). Kelmer–Yu gave an analogue for the sphere. Our proof used dynamics.

References